

LETTERS—CONCISE RESEARCH REPORTS

A Virtual-First Telehealth Treatment Model for Opioid Use Disorder



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INTRODUCTION

Death rates from opioid overdose continue to rise at an alarming pace. Many overdose victims die with untreated opioid use disorder (OUD), despite the availability of effective and lifesaving medications for opioid use disorder (MOUD). Significant barriers to treatment access include transportation, competing responsibilities, stigma, and related costs¹. While millions of individuals receive buprenorphine in a given year², the majority discontinue treatment within a few weeks or months^{3,4} forfeiting ongoing protective effects⁵.

OUD treatment via virtual care platforms may help overcome these common barriers to retention in care^{3,4}. Until recently, the 2008 Ryan Haight Act required an in-person assessment before prescribing any controlled substance, greatly limiting telehealth for OUD¹. However, COVID-19-related reforms largely waived such requirements and hold promise for improving access to care, especially in underserved rural areas. These regulatory changes have attracted innovation among startups as well as incumbent health systems¹. Given long-standing challenges in retaining patients in traditional office-based buprenorphine treatment^{3,4}, research is needed to determine how virtual care outcomes can optimize retention.

METHODS

We analyzed data from a cohort of individuals with OUD treated at Ophelia, a virtual-first telehealth OUD treatment platform. Individuals in New York and Pennsylvania were recruited directly online (e.g., Facebook and Google ads). Medical visits and clinically indicated urine drug screens, organizational structure, custom EHR, and care coordination services were all built explicitly for remote care without

requiring any in-person visits. In-network patients, predominantly Medicaid beneficiaries, used insurance. Patients out of network or uninsured paid \$195 monthly to receive unlimited real-time, video-based clinical visits. Eligible patients (i.e., those with OUD not requiring a higher level of care) were prescribed buprenorphine at intake, and seen weekly during the stabilization phase, and then stepped down to monthly visits under a nurse care manager model.

To investigate 180-day treatment retention, a minimum duration of pharmacotherapy for OUD endorsed by the National Quality Forum⁶, we analyzed a sample of consecutive new intakes from July 1, 2020, to April 15, 2021. Kaplan-Meier survival analyses determined retention, with discontinuation being defined as a 60+ day gap between clinical visits. Consistent with prior studies, care episodes of ≤ 7 days (9% of total episodes) were excluded³ as they often reflected patients ineligible or unwilling to initiate treatment. Geographic heat maps compared distribution of patients to the SAMHSA locator for buprenorphine x-waivered prescribers at the zip code level with eSpatial mapping technology. Patients' home addresses were categorized under USDA Rural-Urban Commuting Area Codes, RUCA codes, with 1–3 denoting urban and 4–10 denoting rural locations.⁷ Secondary outcomes measuring adherence included the proportion of days covered (PDC) and medication possession ratio (MPR) of buprenorphine. The Western (WCG) IRB approved a waiver of consent for the study conducted under STROBE guidelines.

RESULTS

A total of 475 patients were included, 60.3% male, mean age 36.3 years (SD=7.1 years). Two-thirds (66.5%) self-reported race/ethnicity: 88.3% were white. The majority reported Medicaid coverage, consistent with prior studies.^{3,4} The 180-day retention was 69.1% (95%CI: 65.0–73.2%) (Fig. 1); 21.9% of patients resided in rural/small town areas reflecting much greater geographic variation than that of x-waivered prescribers (Fig. 2). 28% of urban patients were in a zip code

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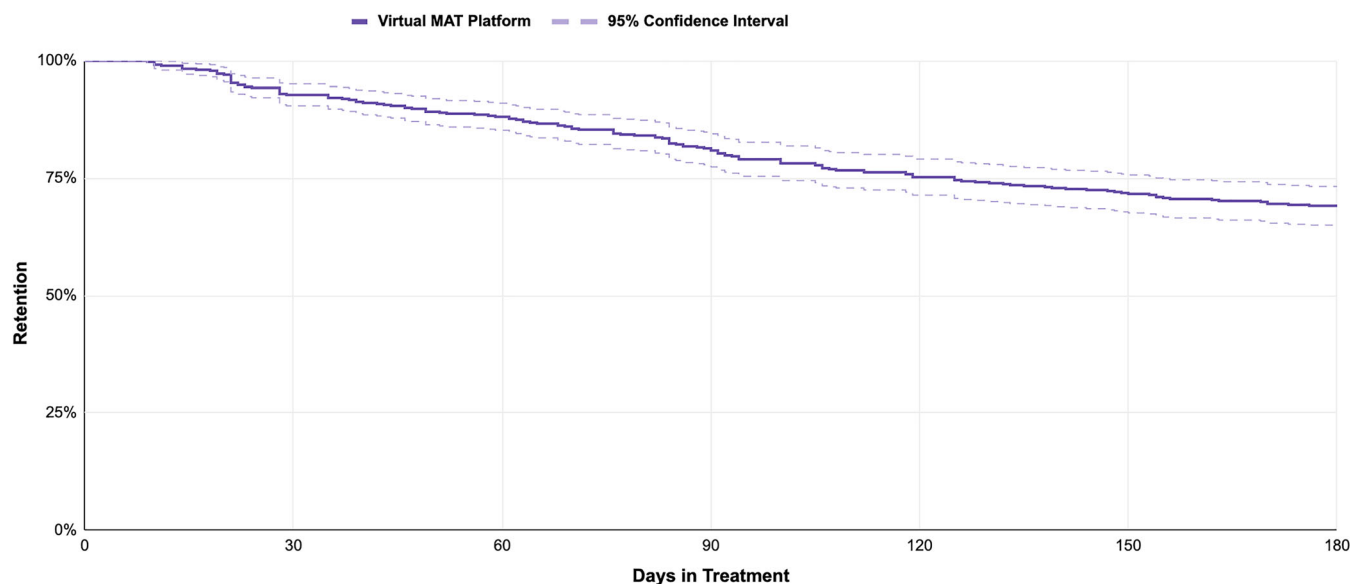


Figure 1 The 180-day retention among patients with opioid use disorder in buprenorphine maintenance treatment provided by telehealth.

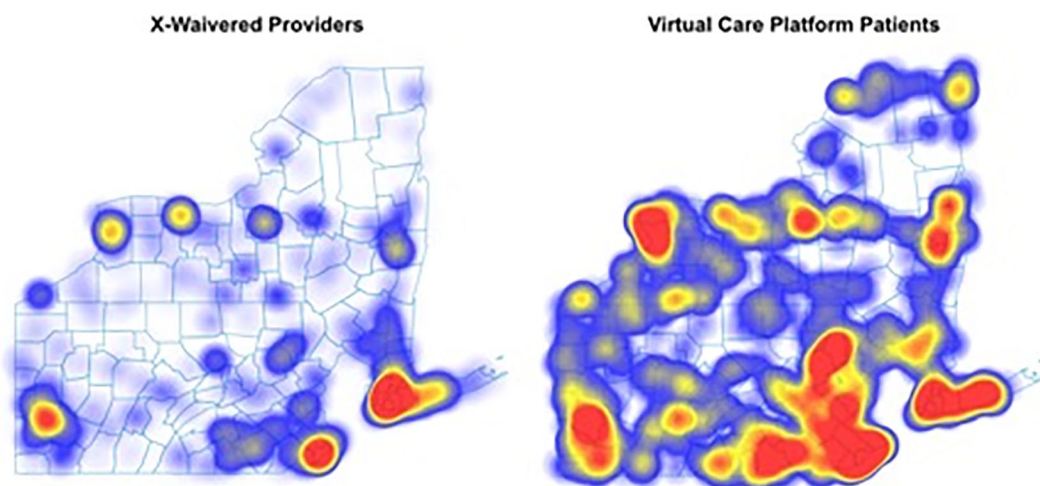


Figure 2 Density distribution of x-waivered buprenorphine providers (per SAMHSA locator) vs patients enrolled via a virtual care platform in Pennsylvania and New York ($n=475$), 2020–2021 data. *Among 475 patients, 43.4% (206) resided in New York and 56.6% (269) in Pennsylvania.

without an x-waivered provider as were 31.96% of rural patients. Patients had a proportion of days covered of 0.96 and a medication possession ratio of 1.05 reflecting high adherence without evidence of stockpiling medication.

DISCUSSION

The observed 180-day retention rate of 69.1% is superior to prior observational studies analyzing multi-state Medicaid (27.0%)³ and commercial insurance prescription claims

(31.0%)⁴. Unlike requirements for in-person care, telehealth enables patients to access care that may not be available locally and attend visits with less interruption and more discretion, decreasing stigma.

Expanding access to medication-based care and improving treatment retention are vital to address the worsening opioid overdose crisis. Technology-enabled telehealth platforms may be important tools for increasing access, retention, and patient satisfaction with evidence-based OUD care. Telehealth is not a single entity and further research is needed to determine best practices, especially for improv-

ing patient retention in OUD treatment with buprenorphine. Further research is also necessary to determine how telehealth interventions can best reach racially and ethnically diverse populations who are underserved by OUD treatment generally and for whom fatal overdose rates are currently rising the fastest.

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Declarations:

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